A single edge density does not determine graph statistics

Conjecture 00000004398

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Scope. We use the printed definition: a forcing graph has a single density determining all graph statistics. Density means homomorphism density. This note refutes the proposed inclusion of all complete graphs under that reading. The Lean theorem concerns finite simple graphs; it does not assert a formalization of graphon integration or asymptotic quasirandomness.

For finite simple graphs F, G with $|V(G)| > 0$, put

$$t(F,G) = \frac{\#\{f : V(F) \to V(G) : f \text{ preserves edges}\}}{|V(G)|^{|V(F)|}}.$$

A necessary consequence of the proposed classification is that equality of $t(K_2, G)$ and $t(K_2, H)$ implies equality of $t(J, G)$ and $t(J, H)$ for every finite simple graph J.

Theorem 1. *The graphs $G = K_3 \sqcup K_1$ and $H = K_{1,3}$ have the same K_2 density and different K_3 densities. Consequently K_2 does not have the stated forcing property, and the proposed classification is false under that definition.*

Proof. Both graphs have four vertices and three edges. Every undirected edge gives exactly two homomorphisms from K_2, so both homomorphism counts are six. A homomorphism from K_3 to a loopless graph must be injective. The triangle of G gives $3! = 6$ homomorphisms; the star H has no triangle and gives none. Thus

$$t(K_2, G) = t(K_2, H) = \frac{6}{16} = \frac{3}{8}, \qquad t(K_3, G) = \frac{6}{64} = \frac{3}{32} \neq 0 = t(K_3, H).$$

Since K_2 is complete, it contradicts the forward implication of the claimed classification. No classification of the remaining graphs is needed. □

Formal correspondence. The file `GraphDensity.lean` defines Boolean adjacency on `Fin n`, simplicity, complete graphs, all vertex maps, the edge-preservation test and homomorphism counts. Density equality uses cross multiplication of positive integer denominators. `mem_allMaps` proves coverage of every vertex map; `enumeration_two` and `enumeration_three` verify sizes 16 and 64 and absence of duplicate tuples in the exact enumerations used here. The final theorem `complete_two_not_forcing` applies the purported property to these actual graphs; `not_all_complete_forcing` refutes the necessary universal implication. Only the standard axiom `propext` appears in the final theorem audit.

Interpretation limit. The usual quasirandom-forcing terminology is more specific than the problem's short definition. This note does not silently replace it with a new asymptotic assertion. The finite computation is exact; acceptance of the printed-definition reading remains for the competition reviewers.

Sources. Official bilingual conjecture is included as `conjecture.md`. For the conventional homomorphism-density definition, see Grzesik, Král' and Pikhurko, Forcing Generalized Quasirandom Graphs Efficiently.